BUSINESS INTELLIGENCE AND TOOLS

EXPERIMENT 10

Report Scheduling, Automation and Alerts using Microsoft Power BI

Stepwise Procedure

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Aim

To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI to ensure availability, reliability, and continuity of BI solutions.

Tools / Software Required

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel
- Microsoft PowerPoint (optional)

Stepwise Procedure

Part A: BI Deployment Planning

1. Identify the BI assets created in the previous experiments, such as reports, dashboards, datasets, and analytical outputs.
2. Prepare an inventory of the reports, dashboards, datasets, and workspaces that need to be deployed.
3. Create a BI Deployment Plan that records each asset, its purpose, its intended audience, and its deployment status.
4. Identify the intended users and departments, for example management, sales, operations, and other relevant business teams.
5. Define deployment objectives and expected outcomes, such as consistent access to reports, timely decision-making, and availability of trusted information.

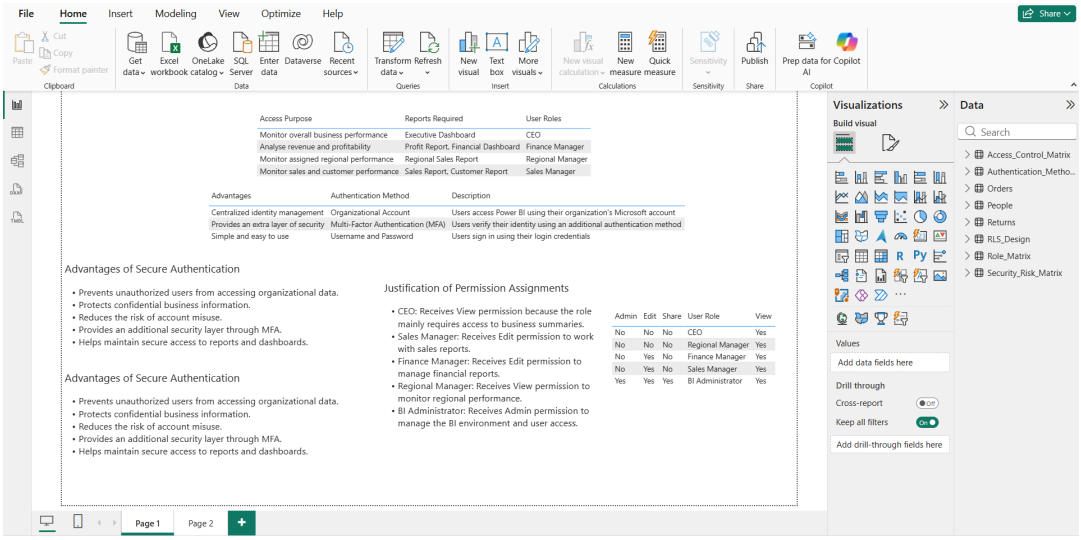


Figure: Page 1 of the Power BI report showing authentication, user roles, and permission planning.

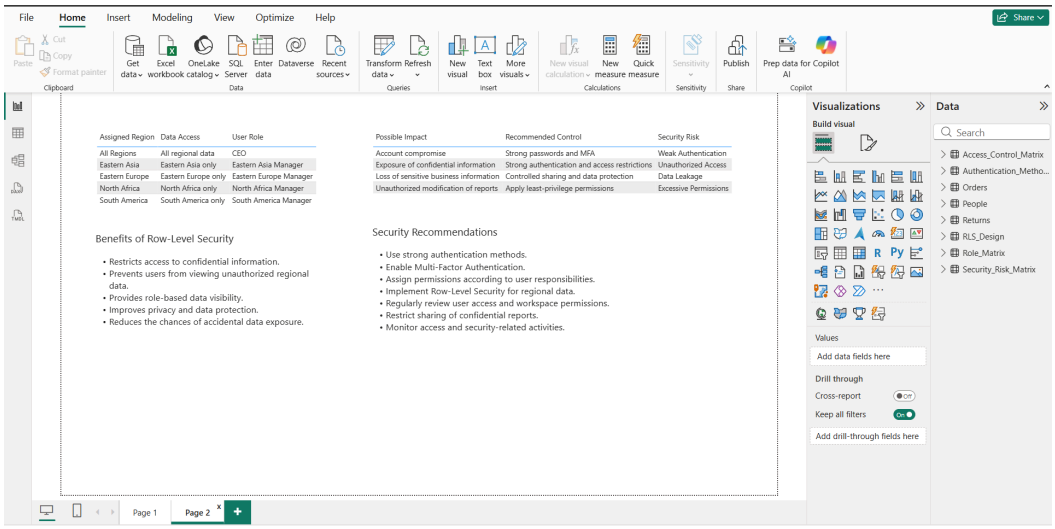


Figure: Page 2 of the Power BI report showing regional data access and security recommendations.

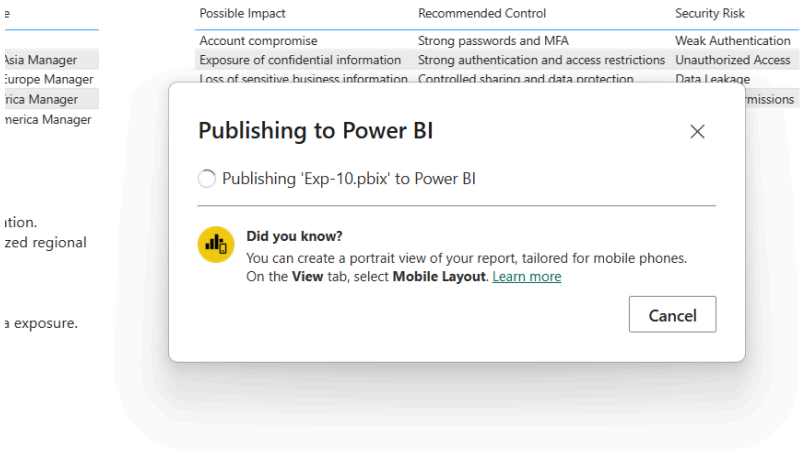


Figure: Power BI Desktop publishing the Exp-10 report to Power BI Service.

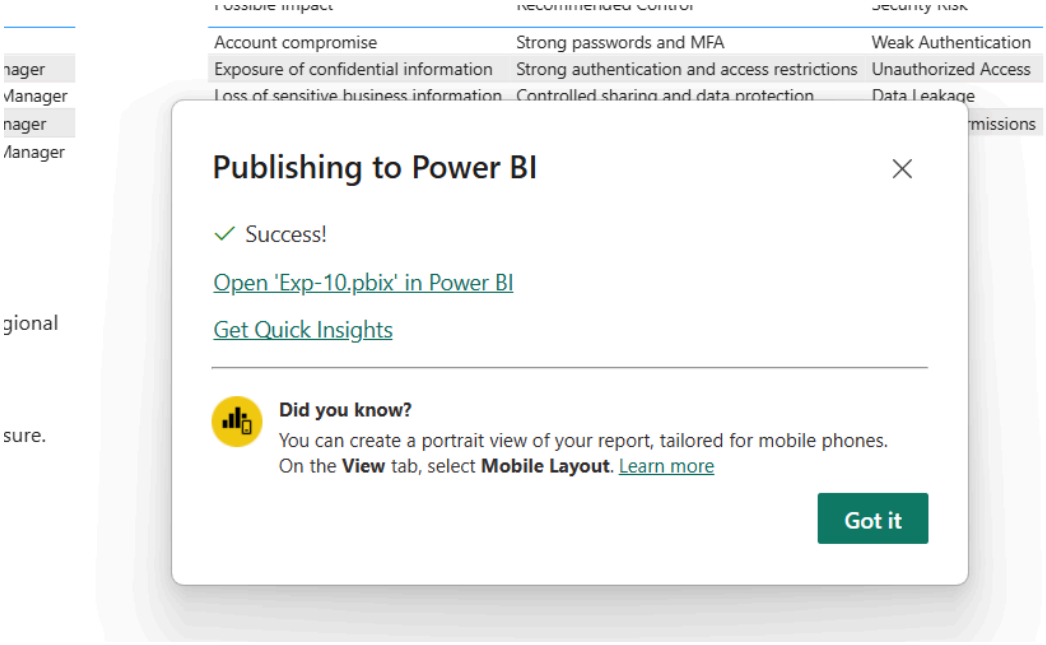


Figure: Successful publication of Exp-10 to Power BI Service.

Part B: Workspace and Access Planning

6. Design a suitable Power BI workspace structure to organize related datasets, reports, and dashboards.
7. Identify the workspace roles: Administrator, Member, Contributor, and Viewer.
8. Assign responsibilities to each role. The Administrator manages workspace settings and access; Members collaborate and manage content; Contributors create and update content; Viewers consume published content.
9. Document workspace governance guidelines covering access approval, naming conventions, ownership, content publishing, and periodic access review.

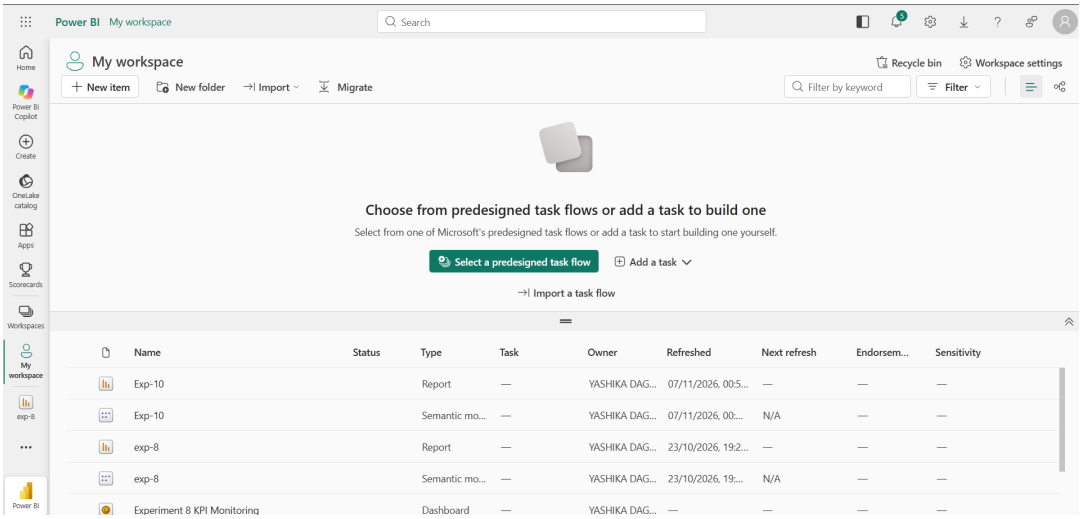


Figure: Exp-10 report and semantic model listed in the Power BI Service workspace.

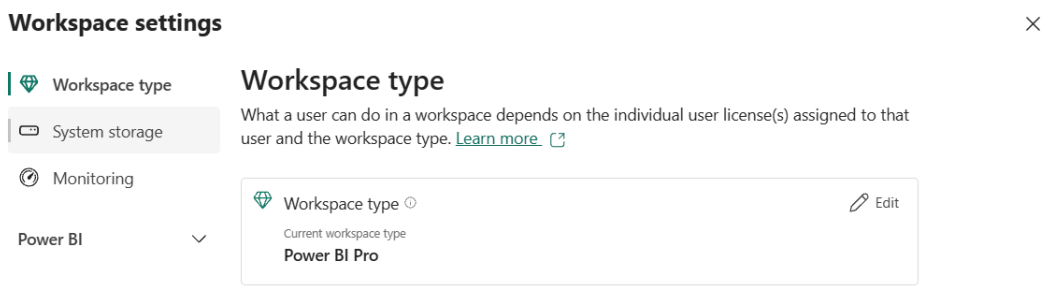


Figure: Workspace settings showing the current workspace type as Power BI Pro.

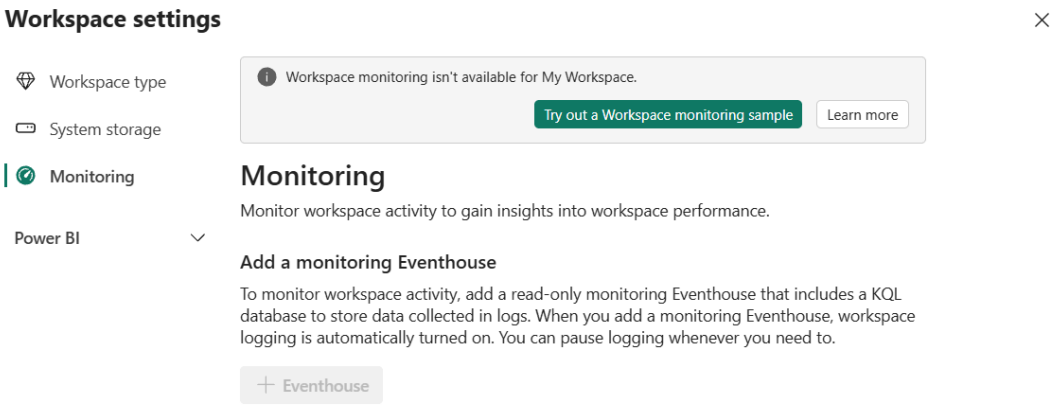


Figure: Monitoring settings page; workspace monitoring is not available for My Workspace.

Part C: Monitoring Strategy Design

10. Identify the monitoring requirements for the deployed BI solution.
11. Include dataset refresh status, dashboard usage statistics, report performance, and user activity monitoring.
12. Prepare a Monitoring Matrix containing each metric, how frequently it will be checked, and the personnel responsible for monitoring it.
13. Define an escalation process for failed refreshes, slow reports, unusual usage patterns, or access issues.

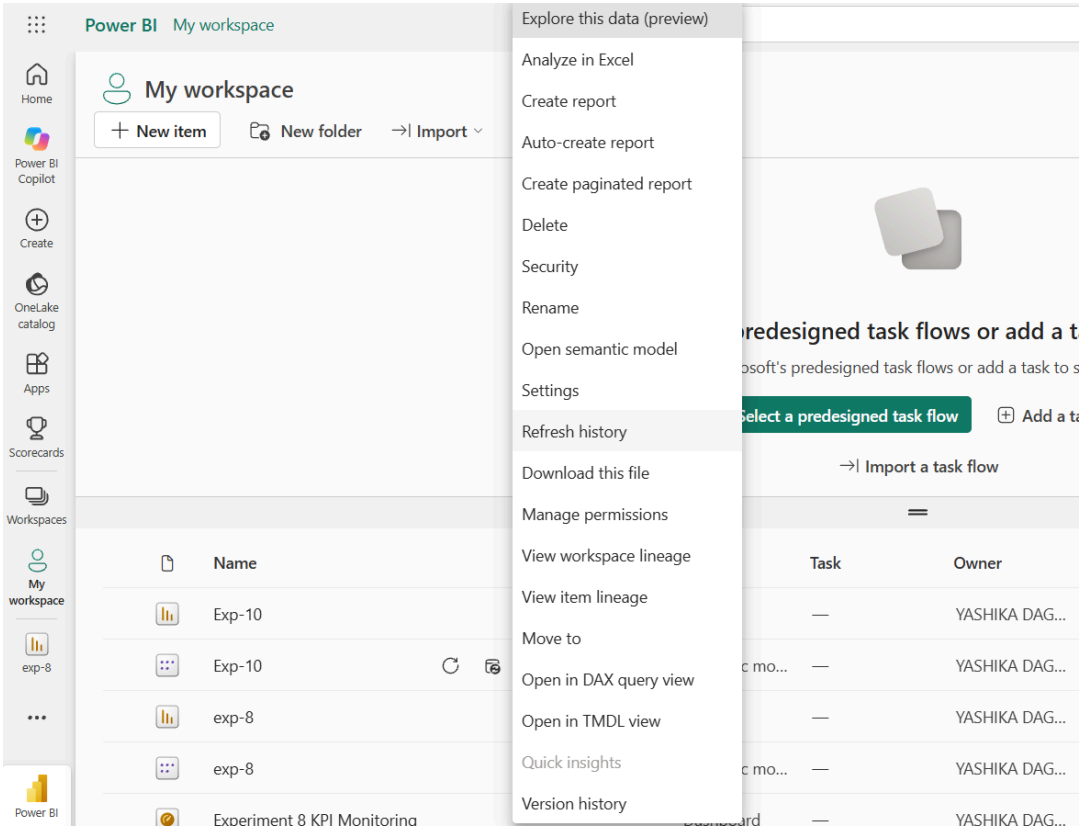


Figure: Refresh history option available from the Power BI Service item menu.

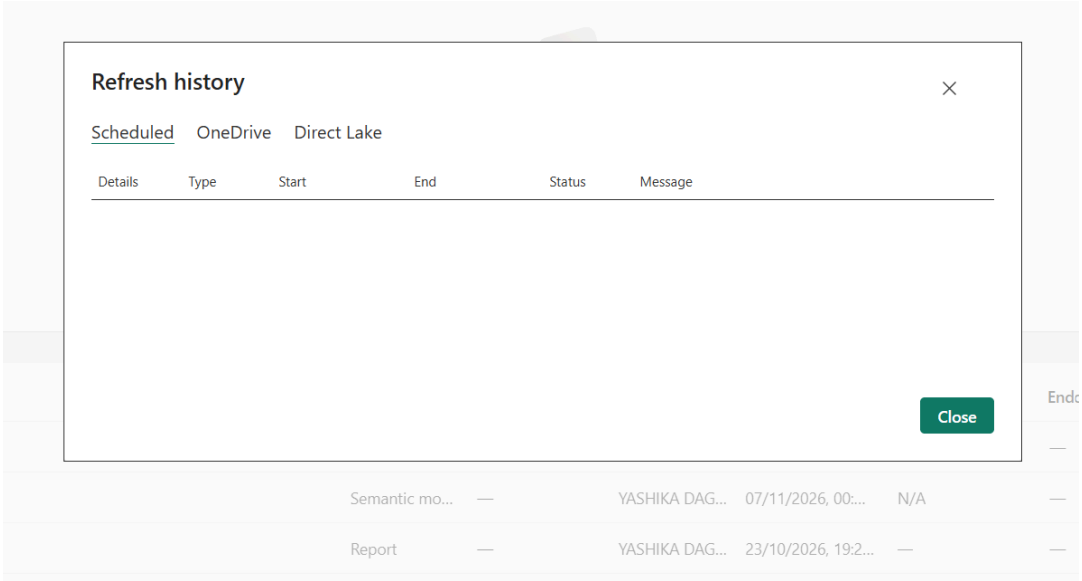


Figure: Refresh history view before a refresh record is displayed.

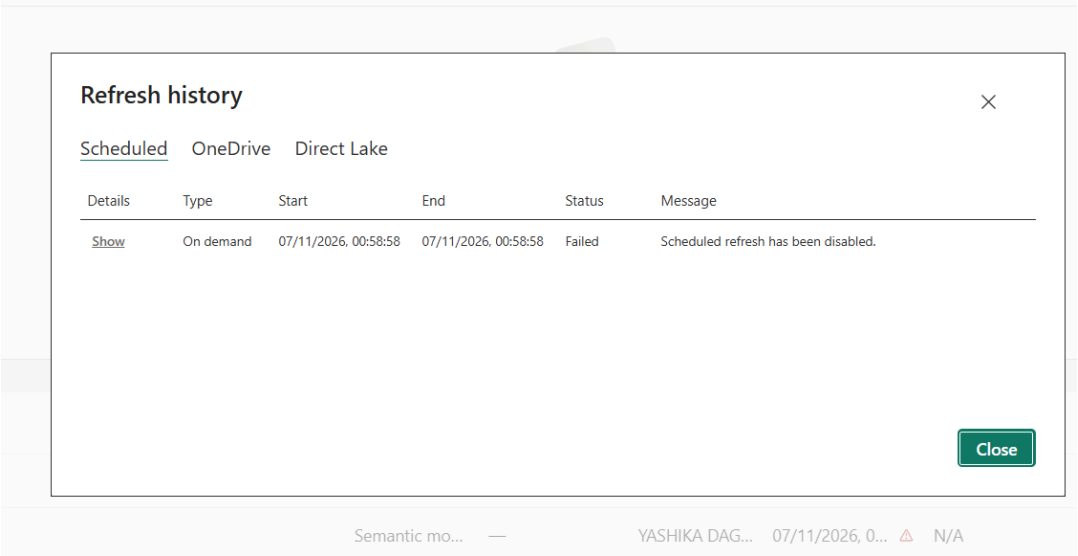


Figure: Refresh history indicating that scheduled refresh has been disabled.

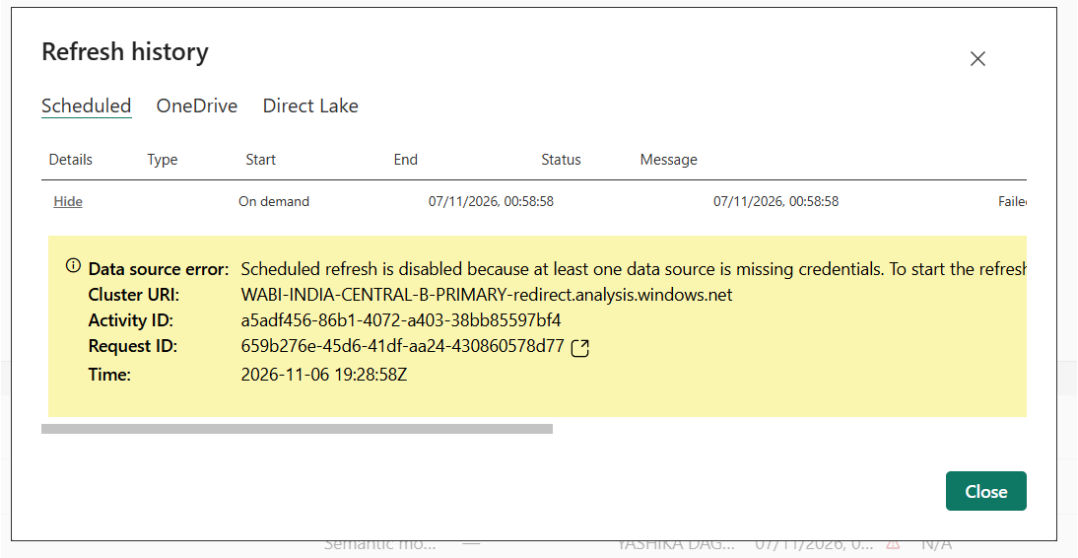


Figure: Refresh failure details identifying missing data-source credentials.

Part D: Backup and Restore Planning

14. Identify the BI components that require backup, including datasets or source files, report files, and workspace-related documentation/configuration.
15. Develop a backup schedule. The experiment suggests daily dataset backup, weekly report backup, and monthly workspace archive.
16. Record the backup owner, storage location, backup date, and verification status for each backup.
17. Design a restore procedure for accidental deletion or system failure: identify the affected asset, locate the latest valid backup, restore or republish the asset, verify data and report functionality, and communicate service restoration.
18. Document the recovery responsibilities and the checks required before declaring the BI solution operational.

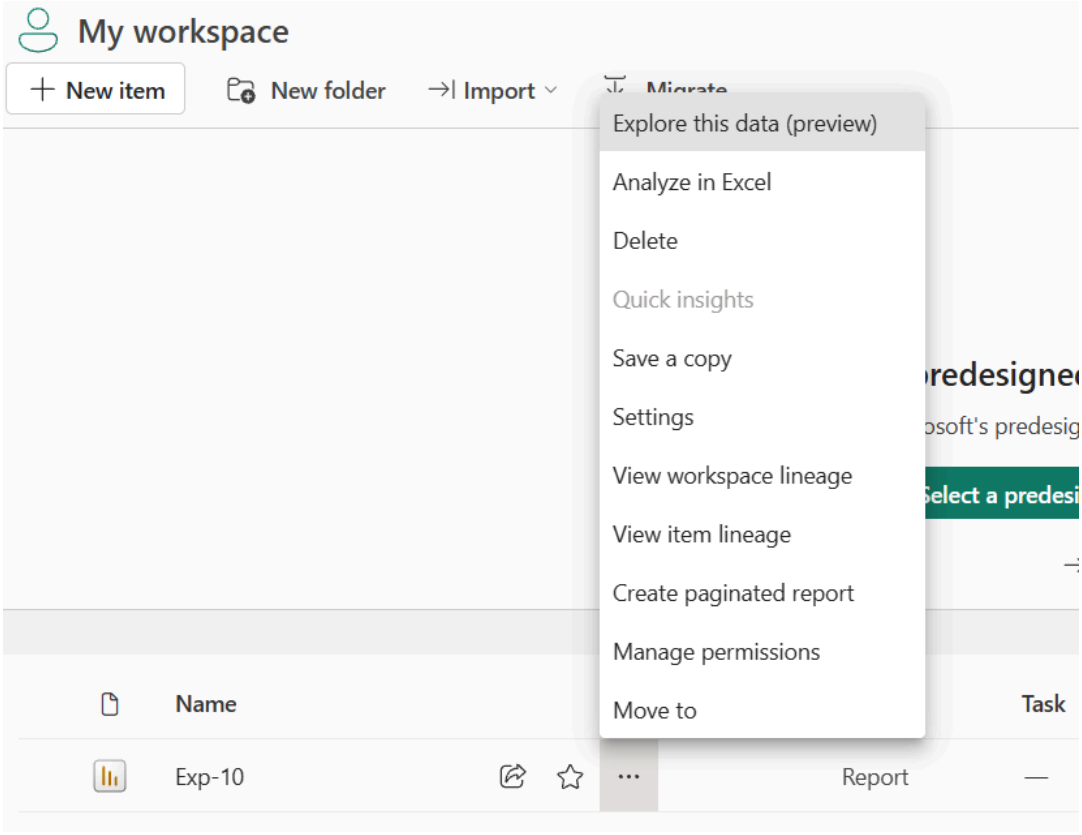


Figure: Power BI Service item menu showing the Save a copy option.

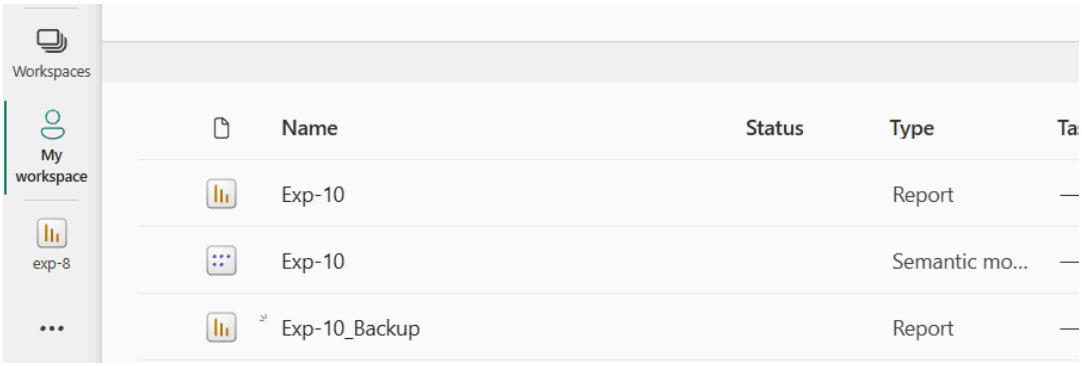


Figure: Backup copy of the Exp-10 report appearing in the workspace.

Part E: Disaster Recovery Scenario

19. Consider the scenario: a critical executive dashboard is accidentally deleted one day before a management meeting.
20. Identify the deleted dashboard, its workspace, its owner, dependent report/dataset, and the most recent available backup or source PBIX file.
21. Notify the BI administrator and relevant stakeholders, and restrict unnecessary changes to the affected workspace during recovery.
22. Restore or republish the dashboard/report from the latest verified backup or retained project file, reconnecting it to the correct dataset if required.

23. Validate that visuals, filters, measures, data refresh, permissions, and dashboard links work as expected.
24. Inform management when access is restored and retain a record of the incident, recovery actions, and preventive improvements.

Result

A BI deployment, monitoring, backup, and recovery strategy was designed for the Global Superstore scenario. The plan covers deployment assets and users, workspace roles and governance, monitoring metrics, backup scheduling, and recovery actions for a critical dashboard deletion.